



# Product Requirements Document

## Deuxième Souffle — APA Coaching Coordination Platform for EHPADs

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| <b>Project</b>      | Deuxième Souffle — APA Coaching Coordination Platform for EHPADs |
| <b>Version</b>      | v0.2 — Draft                                                     |
| <b>Date</b>         | May 21, 2026                                                     |
| <b>Lead PM</b>      | Baptiste Liscouët                                                |
| <b>Secondary PM</b> | TBD                                                              |
| <b>Status</b>       | Discovery Phase                                                  |
| <b>Last Updated</b> | May 21, 2026                                                     |

## 1. Executive Summary

This document is the Product Requirements Document (PRD) for Deuxième Souffle — a France-based platform that coordinates Adapted Physical Activity (APA) coaching sessions delivered by independent APA coaches in EHPAD facilities and similar establishments. The platform replaces an existing Bubble no-code prototype with a purpose-built, production-grade system connecting three user groups: APA coaches, EHPAD facility managers, and the Deuxième Souffle (DS) administrative team. The MVP covers session planning, intelligent coach assignment, geolocated check-in, structured post-session reporting, coach evaluation, contract management, and automated monthly invoicing via Pennylane. It is delivered as a mobile app for coaches, a responsive web client space for EHPAD managers, and a web back-office for the DS team.

### Product Vision

Deuxième Souffle digitises the coordination of Adapted Physical Activity in elderly-care facilities — connecting establishments that need regular sessions with qualified coaches available in their area, and giving the DS team the operational tooling to assign, monitor, and bill those sessions reliably.

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## Primary Objective

Deliver a production-ready MVP covering coach scheduling, the smart matching algorithm, geolocated check-in, the structured post-session report, coach evaluation with configurable coefficients, contract management, and automated monthly billing via Pennylane — replacing the current manual workflow (spreadsheets, phone, email) before national expansion.

## 2. Context & Problem Statement

### 2.1 Business Context

Deuxième Souffle operates as an intermediary between APA coaches and EHPAD facilities, coordinating physical activity sessions for elderly residents. Coordination currently runs through a Bubble no-code application and manual processes (spreadsheets, phone calls, email). As the number of partner EHPADs and contracted coaches grows, these tools have reached their limits in reliability, performance, and feature depth. The decision was made to rebuild the platform on a modern, scalable stack — starting from zero technically, but building on the product vision and business learnings of the existing Bubble application.

### 2.2 User Problem

Coaches lack a reliable mobile tool to view their schedule, position themselves on available sessions, validate attendance, and submit post-session reports. EHPAD managers have no self-service portal to schedule sessions, evaluate coaches, or track invoices. The DS team spends excessive time on manual coach assignment, absence handling, incident management, and invoice generation. The current Bubble platform cannot support geolocated check-in, a composite-score matching algorithm, configurable evaluation criteria, or automated billing. All three user groups are non-technical — the platform must deliver an ultra-simple UX that requires no training.

### 2.3 Opportunity

A purpose-built platform removes the operational ceiling on growth. The matching algorithm scales coach assignment without a proportional increase in administrative effort; geolocated check-in secures session validation and billing; and the Pennylane integration eliminates manual invoicing. Together they turn coordination from a manual bottleneck into a repeatable, auditable process ready for national expansion.

## 3. Target Users & Personas

### 3.1 Primary Persona — APA Coach

Role: APA Coach (independent field professional). Platform: iOS & Android mobile app.

Goals:

- See the weekly and monthly schedule at a glance
- Position on available sessions matching declared availability and service area
- Check in on site via geolocation and submit the structured post-session report quickly
- Track projected and actual revenue

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**Pain Points:**

- No reliable schedule view; assignments communicated informally by phone and email
- Attendance and reporting handled on paper
- Poor visibility on revenue and completed sessions

## 3.2 Secondary Persona — EHPAD Manager

Role: EHPAD Manager (facility coordinator). Platform: responsive web client space (V1 web only; mobile app planned for V2).

**Goals:**

- Schedule regular and one-off APA sessions
- Track upcoming and past sessions and consult coach reports
- Evaluate coaches and manage facility contacts and contracts
- Access and download invoices

**Pain Points:**

- No self-service scheduling tool
- No visibility on which coach is assigned to a session
- Coach quality feedback handled informally
- Invoices received by email with no consolidated history

## 3.3 Tertiary Persona — DS Administrator

Role: DS Administrator (operations team). Platform: web back-office.

**Goals:**

- Assign coaches via the matching algorithm with a full audit trail
- Manage establishments, contracts, and coach profiles
- Handle incidents, absences, and replacements
- Monitor operational and business KPIs and trigger automated monthly invoicing

**Pain Points:**

- Manual, time-consuming coach assignment
- Absence and no-show handling done ad hoc
- Manual, error-prone invoice generation
- No consolidated KPI view of operations

# 4. Feature Scope

The Deuxième Souffle MVP is organised into the feature domains below. Each domain groups multiple user stories; the full, prioritised backlog of 143 user stories across 16 epics is maintained in the project spreadsheet (see Appendices).

- Authentication & Accounts — coach self-registration (email / Google) with admin validation, EHPAD invite-link activation, admin and role management.
- Coach Planning & Check-in — weekly and monthly schedule, available-session positioning, availability declaration, geolocated check-in, Google Calendar sync.
- Smart Assignment — composite-score matching algorithm, attribution calendar, manual override with audit trail, emergency mode, configurable criteria.
- Sessions & Reports — structured 6-step post-session report, inter-session transmission notes, incident management, session-status reference.
- Contracts — regular and one-off scheduling, suggested time slots, automatic session generation, contract lifecycle (create / edit / extend / cancel / approve).

- Billing & Pennylane — automated monthly invoice generation, invoice dashboard and adjustment, Pennylane synchronisation, coach financial dashboard.
- Establishments & Groups — establishment management, contact management, facility-group modelling, rate management.
- Coach Management & Evaluation — coach profiles, confidence index, configurable evaluation coefficients, gamification (badges and levels).
- Notifications & Alerts — session reminders, assignment and delay alerts, no-show detection, configurable notification templates.
- Admin Dashboard & KPIs — operational-health KPIs, activity and revenue, pricing analytics, geographic-coverage and coach-concentration views.

Priorities follow the MoSCoW method. The matching algorithm, geolocated check-in, the 6-step post-session report, and automated Pennylane billing are Must-have for the MVP; gamification, geographic-coverage analytics, and the EHPAD mobile app are deferred to later iterations.

## 5. Technical Architecture & Stacks

TO BE COMPLETED / VALIDATED BY TECH LEADS

### 5.1 Architecture Overview

Deuxième Souffle is a multi-platform system with three front-ends over a shared backend:

- APA Coach — native mobile app (iOS & Android)
- EHPAD Manager — responsive web client space
- DS Administrator — web back-office

Core operational flow:

1. An EHPAD or the DS admin creates a contract; sessions are generated automatically from its recurrence rules.
2. The matching algorithm ranks eligible coaches for each session on a composite score (self-positioning, equity, reliability, proximity).
3. The DS admin validates the suggested coach (pre-assignment mode); the coach is notified and the session syncs to Google Calendar.
4. Coaches can also position themselves on available sessions by raising their hand.
5. On the session day, the coach checks in via geolocation within a configurable time window and radius; the session status moves to In Progress.
6. After the session, the coach submits the structured 6-step report; the status moves to Completed.
7. The EHPAD evaluates the coach; ratings feed the coach confidence index.
8. On the first of each month, the system generates one draft invoice per facility from completed sessions and pushes it to Pennylane.
9. Throughout, the DS back-office monitors sessions, incidents, and KPIs, and configures business parameters.

### 5.2 Detailed Tech Stack

| Layer    | Technology | Rationale                 |
|----------|------------|---------------------------|
| Monorepo | Turborepo  | Single repository sharing |

|                                       |                                    |                                                                                             |
|---------------------------------------|------------------------------------|---------------------------------------------------------------------------------------------|
|                                       |                                    | code across the coach app, EHPAD web space, and admin back-office via packages/shared       |
| Coach app                             | React Native (iOS + Android)       | Cross-platform mobile app for APA coaches                                                   |
| Web (EHPAD space + Admin back-office) | React                              | Responsive EHPAD client space and web back-office                                           |
| Backend / BaaS                        | Supabase (self-hosted)             | Postgres database, authentication, storage, and realtime — self-hosted per DU standards     |
| Runtime                               | Bun                                | Server runtime and tooling                                                                  |
| Database                              | PostgreSQL (via Supabase)          | Users, coaches, establishments, contracts, sessions, reports, invoices, audit logs          |
| Auth                                  | Supabase Auth + Google OAuth       | Email/password and Google social login (coaches); role-based access (Coach / EHPAD / Admin) |
| Storage                               | Supabase Storage                   | Coach KYC documents, avatars, and report attachments                                        |
| Maps / Geolocation                    | DU Maps                            | Address autocomplete and distance / travel-time computation for matching and check-in       |
| CI/CD & Testing                       | GitLab CI — unit, integration, E2E | Automated build, test, and deploy pipelines; per-sprint QA                                  |

### 5.3 APIs & Third-Party Integrations

| Service   | Description                                             | Criticality        |
|-----------|---------------------------------------------------------|--------------------|
| Pennylane | Automated monthly invoice generation and payment-status | Critical — billing |

|                   |                                                                                                             |                            |
|-------------------|-------------------------------------------------------------------------------------------------------------|----------------------------|
|                   | synchronisation                                                                                             |                            |
| Google Calendar   | Unidirectional sync of assigned sessions to coach calendars                                                 | High — coach UX            |
| DU Maps           | Address autocomplete and distance / travel-time computation for the matching engine and geolocated check-in | Critical — matching engine |
| Brevo             | Transactional email (account confirmations, EHPAD invitations, alerts)                                      | High — communications      |
| Telnyx            | SMS notifications                                                                                           | Medium — notifications     |
| Push (FCM / APNs) | Mobile push notifications for session events (assignment, reminder, delay, no-show)                         | High — coach UX            |

## 5.4 Non-Functional Requirements

| Dimension     | MVP Target                                                                                                           | Long-Term Target                                                                 |
|---------------|----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| Performance   | Schedule and matching views load < 2s; matching suggestion computed within seconds                                   | API p95 < 200ms under sustained load                                             |
| Scalability   | France launch; hundreds of active coaches and thousands of monthly sessions                                          | National scale; horizontal auto-scaling                                          |
| Security      | HTTPS end-to-end; role-based access control (Coach / EHPAD / Admin); GDPR-compliant; KYC documents encrypted at rest | OWASP Top 10 compliance; annual security audit; GDPR DPA with all sub-processors |
| Availability  | 99% uptime (backend API + matching engine)                                                                           | 99.9% uptime SLA; automated health checks and alerting                           |
| Accessibility | Ultra-simple UX requiring no training; usable by                                                                     | WCAG 2.1 AA compliance; internationalisation (FR / EN)                           |

|  |                                 |  |
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|  | non-technical users of all ages |  |
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## 6. Roadmap & Planning

### 6.1 Project Phases

| Phase                         | Duration   | Team                                          | Deliverables                                                                                                                                                                                                                                                                                                                                                       |
|-------------------------------|------------|-----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase 0 — Discovery & Scoping | ~1.5 weeks | PM, Tech Lead, UX Designer                    | Scoping workshops; MVP validation; business rules (matching logic, check-in window, cancellation policy, incident handling); prioritised user stories (Coach / EHPAD / Admin); Jira backlog and sprint setup; Work Breakdown Structure and project charter.                                                                                                        |
| Phase 1 — Design & MVP Build  | ~10 weeks  | UX Designer, Mobile Dev, Web Dev, Backend Dev | UX/UI (Figma audit, MVP screens, design system); coach mobile app; EHPAD responsive client space; DS web back-office; Supabase backend; composite-score matching algorithm; geolocated check-in; 6-step post-session report; contract and session generation; Pennylane invoicing; notifications; admin dashboard and KPIs. Deliverable: full staging environment. |
| Phase 2 — Beta & QA           | ~1 week    | QA, Mobile Dev, Web Dev, Backend Dev          | QA of all critical journeys (scheduling,                                                                                                                                                                                                                                                                                                                           |

|                  |         |                     |                                                                                                                                                          |
|------------------|---------|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
|                  |         |                     | matching, check-in, reporting, evaluation, billing); multi-device testing on iOS and Android; pre-production release candidate validated by the client.  |
| Phase 3 — Launch | ~1 week | DevOps, PM, DS team | App Store and Google Play publication; production deployment; monitoring live; DS team handover (runbook, back-office training, corrective maintenance). |

## 7. Appendices & Related Documents

The following documents complement this PRD and were used as input for the project scope:

- User Story Backlog — project spreadsheet listing the full, prioritised set of 143 user stories across 16 epics (English tab '2\_Stories'; French tab '2\_Stories\_FR').
- Work Breakdown Structure (WBS) — epic and task breakdown organised by delivery phase.
- Scoping Workshop & Q&A notes — decisions and business-rule clarifications from the cadrage workshops (matching logic, check-in window, cancellation policy, session statuses).
- Functional specifications and project brief — business objectives, target users, and feature detail.
- Design brief & Figma references — visual identity and MVP screen designs.
- Kick-off presentation — project methodology, tooling (Slack, Jira), and delivery roadmap.

## Version History

| Version | Date           | Author            | Changes                                                     |
|---------|----------------|-------------------|-------------------------------------------------------------|
| v0.1    | March 11, 2026 | Baptiste Liscouët | Initial document creation.                                  |
| v0.2    | May 21, 2026   | Baptiste Liscouët | Full rewrite aligned to the Deuxième Souffle APA-coaching / |

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|--|--|--|----------------------------------------------------------------|
|  |  |  | EHPAD scope and the consolidated 143-story user-story backlog. |
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